

Lakshay Tyagi

Software Developer and Research Engineer building cutting-edge AI solutions at AWS

Building Search, Large Language Model (LLM), Computer Vision, Machine Learning, Retrieval-Augmented Generation, and Distributed Database Solutions

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GitHub | Google Scholar

LinkedIn | Personal Website

Education

New York University, Courant Institute of Mathematical Sciences

MS in Computer Science; GPA: 4.0/4.0; Fellowship for exceptional academic performance

Sep 2022 - May 2024

New York City, NY

Indian Institute of Technology, Kanpur (IITK)

BTech, Double Major Electrical Engineering; GPA: 9.5/10.0; General Proficiency Medal

Jul 2017 - May 2022

Kanpur, UP

Work Experience

Amazon Web Services (AWS) - Software Development Engineer - AI/ML

Jul 2024 - Present

New York City, NY

- Architected integration for custom Machine Learning models in Bedrock Knowledge Bases, improving accuracy by 20%.
- Optimized LLM inference infrastructure with vLLM and Text Embeddings Inference (TEI), reducing latency by 10% and improving throughput by 20%.
- Implemented Agentic Retrieval with inferred actions, leading to a 10% improvement in LLM response accuracy.
- Improved retrieval for hierarchical chunking, storing text in a NoSQL database, and reducing latency by 200 ms.
- Optimized Elasticsearch clusters for enterprise search, reducing latency by 47% and saving over \$150,000 per year.
- Revamped CloudFormation CI/CD pipelines to reduce operations overhead by 40% and improve deployment frequency.

NYU Langone Institute - Research Engineer - Publication

Oct 2023 - May 2024

New York City, NY

- Developed Large Language Model (LLM) solutions for labeling medical reports with pathologies and explanations.
- Pretrained BERT-based models on a corpus of 3,400,000 medical reports with Masked Language Modeling.
- Engineered Coreset-based Active Learning algorithms to select 11,000 reports for labeling with a teacher model.
- Trained LLaMA 2 using LoRA with distributed GPU training in PyTorch to improve Macro F1 score to 0.89.

Internships

Bayer, AI Team - Data Science Intern

Jun 2023 - Sep 2023

Whippany, NJ

- Orchestrated time-series forecasting pipelines in Databricks for predicting sales growth, integrated with Snowflake using SQL.
- Engineered features and trained XGBoost, LSTM (RNN), and graph models to reduce MAPE to 0.11 on backtests.
- Developed a Flask web application for visualizing forecasts for over \$100 million in sales across 20 product segments.

Mitacs Globalink - Research Intern - Publication

Jun 2021 - Aug 2021

Calgary, Canada

- Set up a distributed training framework in PyTorch for UNet-based federated learning models for brain tumor segmentation.
- Achieved Dice Similarity Coefficient of 0.674 using variable local tuning with learning rate scheduling.

Samsung Research Institute - Software Engineering Intern

May 2020 - Jul 2020

Noida, India

- Implemented autoencoder-based deep learning models in TensorFlow for denoising low-light, grainy videos.
- Delivered state-of-the-art SSIM of 0.89 and PSNR of 30.14 for video generation using perceptual loss and sensor input.

Technical Skills

Programming Languages: Python, Java, C/C++, Ruby, Scala, TypeScript; Databases: DynamoDB, Elasticsearch, Snowflake, Postgres

Cloud Technologies: AWS, Azure, Databricks, Kubernetes, Hadoop, Spark, Grafana; Software: MATLAB, R

Libraries: PyTorch, TensorFlow, Keras, NumPy, Scikit-Learn, OpenCV, pandas, HuggingFace, vLLM, TEI, Spring/Boot, cvxpy, NLTK

Publications

- Humor@IITK at SemEval-2021 Task 7: Large Language Models for Quantifying Humor and Offensiveness. A Gupta*, L Tyagi*, A Pal*, B Khurana*, A Modi: Publication in Association of Computational Linguistics (ACL), ACL-IJCNLP 2021.
- Federated Learning Using Variable Local Training for Brain Tumor Segmentation: A. Tuladhar, L. Tyagi, R. Souza, N.D. Forkert, BrainLes MICCAI 2021.